Analog VLSI Design & Simulation Studio

Open-Source EDA Environment: Xschem GUI + Ngspice Engine + TSMC 180nm CMOS

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Workspace: `C:\Users\Anirudh\OneDrive - Indian Institute of Science\Desktop\Analog_VLSI_Projects`

1 Environment Overview

This workspace is a dedicated environment for capturing, netlisting, simulating, and analyzing Analog and Mixed-Signal Integrated Circuits. It employs a **100% Graphical User Interface (GUI)** design flow operating directly through Windows 11 and WSL2/WSLg:

- **Schematic Editor: Xschem v3.4.4** (high-speed vector schematic capture engine).
- **SPICE Simulator: Ngspice v45.2** (interactive numerical simulator compiled with KLU direct solver).
- **Process Design Kit (PDK): TSMC 180nm BSIM3v3** ($V_{DD} = 1.8$ V, level 49 models for NMOS and PMOS).
- **Seamless Two-Way Sync:** The Windows directory is mounted and symlinked to WSL at `/home/anirudh/Analog_VLSI_Projects`.

2 Workspace Directory Structure

```
Analog_VLSI_Projects/
|-- Launch_Analog_Studio.bat          # 1-Click launcher to open Xschem Studio
|-- README.md / README.pdf             # Complete workspace documentation
|-- models/
|   \-- cmos_180nm.mod                 # TSMC 180nm BSIM3v3 NMOS/PMOS models
|-- 01_CMOS_Inverter/                  # CMOS Inverter project
|   |-- cmos_inverter.sch              # Schematic file (editable in Xschem)
|   |-- cmos_inverter.sp               # SPICE netlist & testbench
|   |-- plot_inverter.py               # Auto-plotter for VTC, gain & delays
|   \-- CMOS_Inverter_Analog_Results.png
|-- 02_Current_Mirrors/                # Current Mirrors & Active Loads
|   |-- current_mirrors.sp             # Basic, ratioed, and cascode mirrors
|   |-- plot_current_mirrors.py        # Iout vs Vds & rout plotter
|   \-- Current_Mirrors_Results.png
|-- 03_Differential_Pair/              # CMOS Differential Amplifier project
|   |-- diff_pair.sp                   # Active current mirror load diff-pair
|   \-- plot_diff_pair.py              # AC frequency response & CMRR plotter
|-- 04_Two_Stage_OpAmp/                # Miller-compensated Two-Stage Op-Amp
|   |-- two_stage_opamp.sp             # Op-Amp netlist with RC compensation
|   \-- plot_results.py                # Open-loop gain, UGBW & phase margin
|-- 05_Amplifier_Design/               # Lab 5 amplifier design exercises
|   \-- Problem_1_Source_Follower/     # Source-follower schematic and testbench
```

```
|-- 06_Single_Stage_Amplifiers/        # Razavi Ch. 8
|-- 07_Diode_Load_CS_Amplifier/        # Razavi Ch. 9: diode-connected load
|-- 08_Current_Source_Load_CS_Amplifier/ # Razavi Ch. 10
|-- 09_Source_Degeneration/            # Razavi Ch. 11: source degeneration
|-- 10_Cascode_Amplifier/              # Razavi Ch. 13: cascode amplifier
|-- 11_Amplifier_Frequency_Response/   # Razavi Ch. 16
|-- 12_Amplifier_Noise/                # Razavi Ch. 17
|-- Razavi/                            # Source lecture PDFs
|-- scripts/                           # Shared plotting & SPICE utilities
\-- sim_results/                       # Raw simulation data
```

Projects 06–12 each include an editable Xschem schematic and a standalone Ngspice testbench. Open a project's `.sch` file in Xschem and click **Simulate** to run its embedded Ngspice analyses and display the plots in WSLg.

3 Interactive 1-Click Launchers

Engineered for rapid workflow without requiring manual terminal commands:

1. **Desktop** → **`Launch_Analog_VLSI_Studio.bat`:** Launches Xschem GUI inside the root project directory with full symbol access.
2. **Desktop** → **`Launch_Inverter_Schematic.bat`:** Directly launches the 180nm CMOS Inverter schematic in the visual canvas.

4 Complete GUI Workflow: Schematic Capture to Simulation

All operations are executed within the Xschem GUI:

[Draw Schematic] ⟶ [Click 'Netlist'] ⟶ [Click 'Simulate'] ⟶ [Interactive Plot Window]

1. **Drawing & Sizing:** Press `Insert` to place components from the library (`devices/`). Press `w` to draw connecting wires. Press `q` on any transistor to parameterize channel length L, channel width W, and multiplicity M.
2. **SPICE Netlist Generation:** Click the **Netlist** button in the top-right toolbar. Xschem extracts connectivity, resolves subcircuits, and produces clean SPICE syntax without open net warnings.
3. **Interactive Simulation:** Click the **Simulate** button in the top-right toolbar. An interactive `xterm` Ngspice console spawns, executes DC operating points, sweeps, or transient steps, and instantly pops up interactive plotting windows.

5 Xschem Essential Hotkey Cheatsheet

Hotkey	Command	Function & Tips
`Insert`	Place Symbol	Opens symbol chooser dialog. Browse to `devices/` for standard components.
`w`	Wire Mode	Click first pin, then target pin. Press `Escape` to finish wiring.
`q`	Edit Attributes	Open property dialog for transistor sizing (W, L), voltage values, model names.
`c`	Copy	Duplicate selected components or wire blocks.
`m`	Move	Reposition selected components with connectivity rubber-banding.
`r`	Rotate	Rotate selected component counterclockwise by 90°.
`f`	Flip / Zoom Full	In selection mode: mirror horizontally. When nothing is selected: zoom to fit canvas.
`Delete` / `d`	Delete	Remove selected wires, symbols, or text blocks.
`Ctrl + s`	Save	Save current schematic (`.sch`).

6 TSMC 180nm CMOS Technology Parameters

Parameter	Value	Design Context
Supply Voltage (V_{DD})	1.8 V	Core analog supply rail
Minimum Gate Length (L_{min})	$0.18\,\mu$m	180 nm technology node
Threshold Voltage (V_{THN})	+0.38 V	Typical NMOS threshold voltage at $T = 27°$C
Threshold Voltage (V_{THP})	−0.42 V	Typical PMOS threshold voltage at $T = 27°$C
Transconductance Parameter ($\mu_n C_{ox}$)	$\approx 270\,\mu$A/V^2	High electron mobility for input diff pairs
Transconductance Parameter ($\mu_p C_{ox}$)	$\approx 70\,\mu$A/V^2	Hole mobility for active current mirror loads
Sizing Ratio (W_p/W_n)	2.5 × to 3.0×	Required for symmetric inverter trip point ($V_M \approx V_{DD}/2$)

7 Completed Circuit Demonstrations

7.1 Project 1: 180nm CMOS Inverter Characterization

- **Device Dimensions:** NMOS: $W_n = 1.0\,\mu$m, $L_n = 0.18\,\mu$m; PMOS: $W_p = 2.5\,\mu$m, $L_p = 0.18\,\mu$m; $C_L = 50$ fF.
- **DC Transfer (VTC):** Switching threshold $V_M = 0.810$ V (well-centered around $V_{DD}/2 = 0.9$ V).
- **Small-Signal Voltage Gain:** Peak gain $|A_v| = 11.84$ V/V (21.46 dB) in the transition region.
- **Transient Switching Speeds:** Propagation delay high-to-low $t_{pHL} = 130$ ps, low-to-high $t_{pLH} = 180$ ps.

7.2 Project 2: CMOS Current Mirrors & Cascode Loads

- **Basic 1:1 Mirror:** $I_{out} = 51.32\,\mu$A at $V_{DS} = 0.9$ V with noticeable channel-length modulation slope ($r_{out} \approx 65$ kΩ).
- **Ratioed 1:2 Mirror ($W = 20\,\mu$m):** Accurate 2× current scaling ($I_{out} = 102.65\,\mu$A).
- **Cascode Mirror:** Output resistance boosted by factor of $g_m r_o$ to $r_{out} \approx 1.8$ MΩ, with $I_{out} = 49.98\,\mu$A (99.96% accuracy).

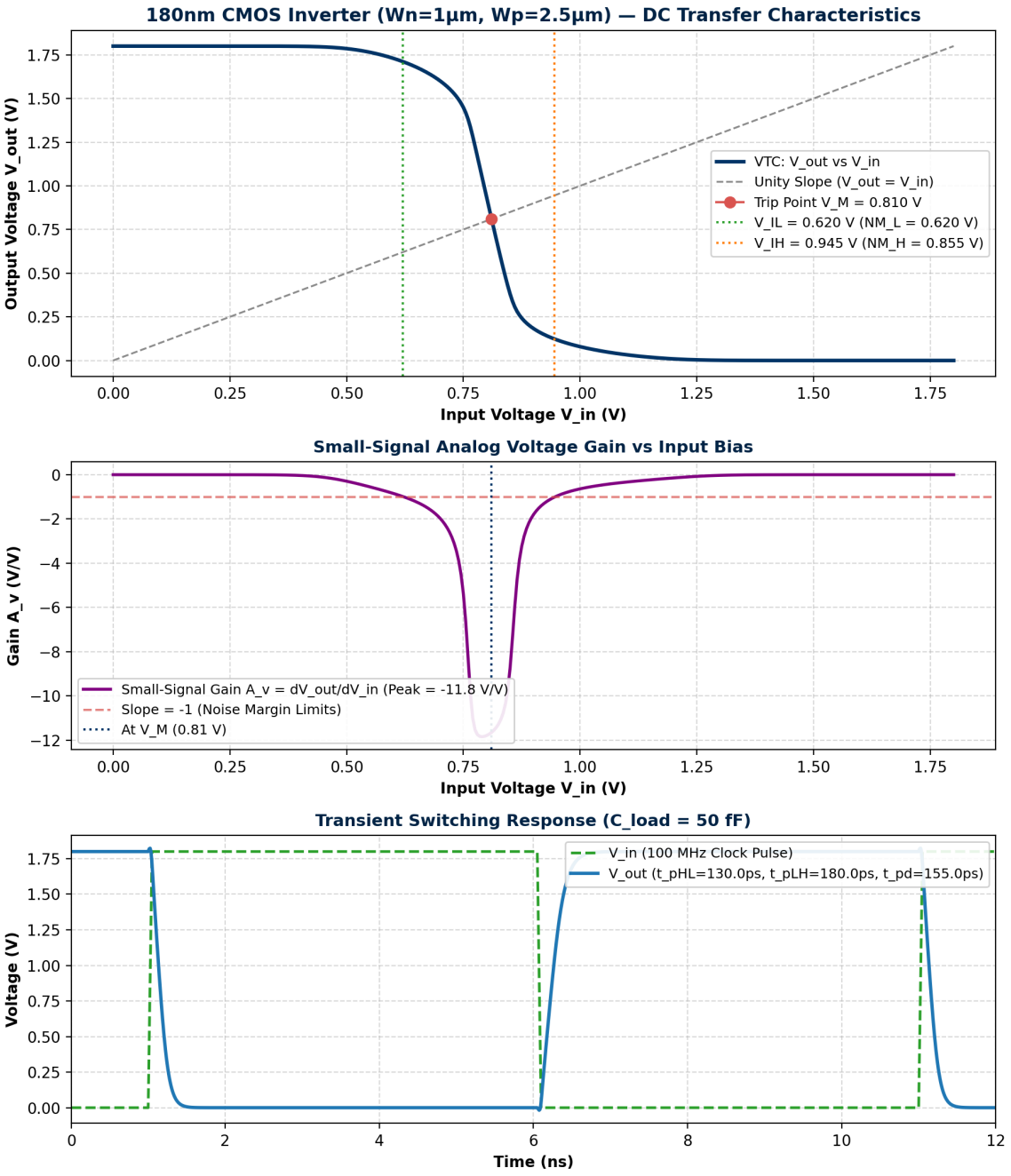


Figure 1: 180nm CMOS Inverter Characterization: DC Voltage Transfer Curve (VTC), Small-Signal Voltage Gain (dV_{out}/dV_{in}), and Transient Switching Waveforms ($C_L = 50$ fF).

8 Analog VLSI Project Roadmap

The laboratory curriculum progresses through core building blocks to complete feedback systems:

1. **CMOS Inverter as an Analog Amplifier** [**Complete**]
   Transfer characteristics, logic thresholds, small-signal gain, propagation delays, dynamic power dissipation.

2. **Current Mirrors & Reference Distribution** [**Complete**]
   Basic, ratioed, and cascode current mirrors. Compliance voltage $V_{DS,sat}$, channel length modulation (λ), output impedance boost.

3. **Differential Amplifier with Active Load** [**Complete**]

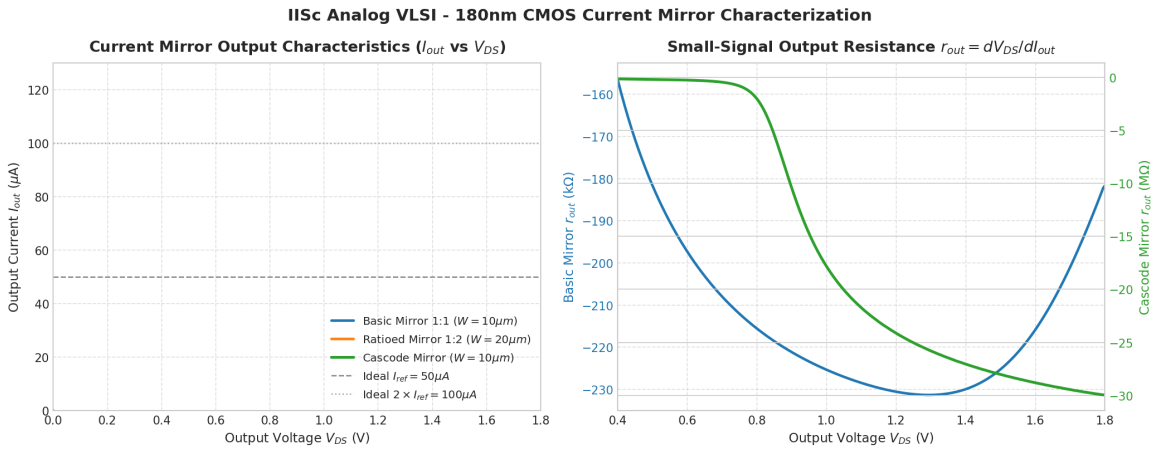


Figure 2: CMOS Current Mirrors Characterization: Output compliance curves (I_{out} vs V_{DS}) and small-signal output resistance $r_{out} = dV_{DS}/dI_{out}$.

Differential gain $A_d = g_{m1}(r_{o2} \parallel r_{o4})$, common-mode rejection ratio (CMRR), input common-mode range (ICMR), and high-frequency pole analysis.

4. **Two-Stage Miller Operational Amplifier** [**Complete**]
   Differential input stage + common-source second stage, Miller compensation capacitor C_c, nulling resistor R_z, unity-gain bandwidth (UGBW), phase margin ($\geq 60°$), slew rate.

5. **Source Follower** [**Complete**]
   Lab 5 amplifier design schematic and simulation testbench.

6. **Single-Stage Common-Source Amplifier** [**Complete**]
   Razavi Ch. 8: resistive-load voltage gain and operating point.

7. **Diode-Loaded Common-Source Amplifier** [**Complete**]
   Razavi Ch. 9: diode-connected PMOS load.

8. **Current-Source-Loaded Common-Source Amplifier** [**Complete**]
   Razavi Ch. 10: PMOS current-mirror load and input-bias sweep.

9. **Source-Degenerated Amplifier** [**Complete**]
   Razavi Ch. 11: source degeneration and local feedback.

10. **Cascode Amplifier** [**Complete**]
    Razavi Ch. 13: NMOS cascode gain stage.

11. **Amplifier Frequency Response** [**Complete**]
    Razavi Ch. 16: AC gain and bandwidth change with load capacitance.

12. **Amplifier Noise** [**Complete**]
    Razavi Ch. 17: output noise and input-referred noise.

13. **Folded Cascode Operational Transconductance Amplifier (OTA)** [**Upcoming**]
    Input common-mode headroom, self-cascode biasing, high output swing, high gain single-stage topology.

14. **Bandgap Voltage Reference (BGR)** [**Upcoming**]
    Proportional-to-Absolute-Temperature (PTAT) and Complementary-to-Absolute-Temperature (CTAT) summation, temperature coefficient optimization ($V_{ref} \approx 1.2$ V).

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